

Precept 1 Outline (KEY)

ECO 202 TAs

September 10, 2026

1 Introductions (5 min)

- TA: name, year, where they are from, what their research is about, and answer a quick icebreaker (TA's choice)
- Students: their name, whether they have taken a stats class before, same icebreaker question as above

2 Overview & expectations for precept (5 min)

1. Purpose of precept:

- We will use precept to run through key practice problems and explain solutions on the board. Our goal will be to focus on statistical concepts themselves (as opposed to other goals of this course, like using AI tools ethically and effectively).
- We will post outlines like this document on GitHub: [Link to folder](#)

2. Tech policy:

- Please, no laptops or phones unless we are doing an activity together and ask you to try it on your computer, or something like this. The only materials you should typically need is something to take handwritten notes with (iPads are fine).

3. If you miss a precept:

- Ask a classmate for notes for reference. You can also try the problems listed on the posted precept document from that week, check what the solutions look like, and come to office hours with any remaining questions.

4. Asking questions has positive externalities!

- No question is a “dumb question.” We are happy to see you asking questions! Plus, it is helpful for you and for anyone else in the class to hear the question and the answer. So, please ask questions often and without hesitation.

5. We as TAs can't be 1:1 tutors, but there are multiple options for extra support:

- Come to office hours for any preceptor, as much as you want.
 - Dexin: Tuesdays, 4-5pm, tables outside of JRR A19
 - Emily: Thursdays, 4:30-5:30pm, JRR A74
 - Ryan: Fridays, 2-3pm, JRR A74
- Come to office hours of the professor: Fridays, 3pm-4pm, JRR 278
- Check out the McGraw Center:
 - Individual tutoring: [Link to page](#)

- Group study with a tutor: [Link to page](#)
 - Reach out to us by email if you’ve tried the above but are seeking support beyond these things.
6. Email us if you have questions that can’t be answered in precept, or in office hours.
 - If you email a TA, we can usually get back to you within 24-48 hours.
 - A response may take longer if you reach out late at night, or over the weekend.
 7. Quick admin points to check in about:
 - Would you be able to attend the adjacent precept time in this block? (Your TA will explain – our sections are currently pretty small, so we are considering consolidating.)
 - If you are requesting an accommodation with ODS, please do that as soon as possible. We need to know about this for scheduling exam rooms, etc.
 8. We are aware this course is experimental in how it is structured:
 - We will periodically check in about how things are going in the class, but please don’t hesitate to ask us questions about the course itself or tell us your thoughts.
 - For example, in office hours we could talk about how to set up your workflow for taking notes & doing practice problems.
 9. Any questions about the course before jumping in to this week’s material?

3 Review of practice questions (40 min)

Main goal today: Practice with concepts related to describing a dataset and describing a distribution. We will also start practicing how to tell the difference between observational and causal evidence (but we will get into this in more detail after Lecture 5).

1. (From Practice 1, Question 4)

A classmate asks you: “Does staying longer with the same employer raise a worker’s hourly wage?”

The same historical **wage1** file used in lecture contains **tenure**, measured in years with the current employer, and **wage**, measured in 1976 dollars per hour.

- (a) Suppose that you created a scatterplot with **tenure** on the x-axis and **wage1** on the y-axis, and that the points on the scatterplot slope upward. Write the strongest one-sentence conclusion supported by that fact alone.
 - “Among the workers recorded in this historical 1976 extract, workers with longer tenure at their current employer tend to have higher recorded hourly wages.”
 - Circle/point out a couple of things:
 - We are not saying anything about longer tenure *causing* higher hourly wages
 - We are making sure our statement is about *workers represented in the dataset only*, not all workers that every existed in any place, at any time.
- (b) Name two alternative explanations for an observed positive relationship and one kind of design information that would be needed before making a causal claim.
 - Example alternative explanations
 - Age (older people have higher tenure and are generally better paid), likewise, labor-market experience
 - Education level (maybe the jobs available to people with higher degrees are generally both more stable, and higher paying), likewise, occupation/employer/job type

- Worker performance (you keep your job when you do well, and you might get a raise or get promoted, etc.)
- As with all of these, individual selection into longer-lasting jobs
- Example design information:
 - We randomized tenure to these workers somehow...or some other research design
 - Again, we will talk more about this post-lecture 5
- Bigger takeaway: notice how our alternative explanations go hand in hand with knowing that we shouldn't make a *causal* statement without more information.

2. (From Practice 2, Question 1)

A city housing survey has one row for each responding renter household. Its columns include `monthly_rent` in dollars per month, `bedrooms`, `neighborhood`, `rent_burdened` coded 1 when rent exceeds 30 percent of household income and 0 otherwise, `response_mode` coded web or telephone, and `survey_weight`.

(a) State the observational unit.

- Obs. unit = one responding renter household.
- We find the observational unit by thinking about what makes a row in the dataset unique.
- Quick comprehension Q: what if this dataset had rows from the same responder renters every month? What would the observational unit be then? (responder-by-month)
- Also, notice: The word “responding” matters: the rows are not automatically all renter households in the city.

(b) Classify each listed variable as categorical or quantitative. Explain why the values 0 and 1 do not make `rent_burdened` a quantitative variable.

Comprehension/participation Q: How do we tell the difference between categorical/quantitative variables? Could write a short summary based on responses on the board.

So, we have:

- One row = 1 responding renter HH
- Dataset columns:
 - `monthly_rent`: \$\$/month (**quantitative**)
 - `bedrooms` (**quantitative**)
 - `neighborhood` (**categorical**)
 - `rent_burdened`: = 1 when rent > 30% HH income, 0 otherwise (**categorical**)
 - `response_mode`: “web” or “telephone” (**categorical**)
 - `survey_weight` (**quantitative**)
 - Obs. unit = one responding renter HH.

Notice that `rent_burdened` is not quantitative. The values 0 and 1 are numeric, but they label *membership* in the two `rent_burdened` categories.

(c) Explain why a difference in rent between the two `rent_burdened` groups would be descriptive rather than evidence that the coding itself caused rent to change.

`rent_burdened` is partly defined using rent relative to income, so a rent difference between the recorded groups may reflect that construction and other household characteristics. The comparison describes the observed groups; it does not represent an intervention or establish a causal effect of applying a label.

3. (From Practice 2, Question 2) The ordered hourly wages, measured in dollars per hour, for six workers are

4, 6, 7, 8, 9, 20

Use the median-of-halves convention for quartiles.

- (a) Before calculating, predict whether the mean will be below, equal to, or above the median. Explain your prediction from the distribution's shape.

Have students think about it for a second and then solicit their guesses.

The mean exceeds the median, as predicted from the long right tail created by 20.

- (b) Calculate the mean, median, Q_1 , Q_3 , IQR , range, sample variance, and sample standard deviation.

The six values sum to 54, so

$$\bar{x} = \frac{54}{6} = 9.$$

For the median, we care about either the literal “middle” number (if the total number of items is odd, divide # of items by 2 and round up), or the average between the numbers that straddle the “middle” (if the total number of items is even, divide # of items by 2 and look on “either side”). So in this case, the median is the mean of the third and fourth ordered values:

$$M = \frac{7 + 8}{2} = 7.5.$$

The lower half is 4, 6, 7 and the upper half is 8, 9, 20. Under the median-of-halves convention,

$$Q_1 = 6, \quad Q_3 = 9, \quad IQR = 9 - 6 = 3,$$

and the range is $20 - 4 = 16$. The mean exceeds the median, as predicted from the long right tail created by 20.

The squared deviations from the mean are

$$(4 - 9)^2, (6 - 9)^2, (7 - 9)^2, (8 - 9)^2, (9 - 9)^2, (20 - 9)^2,$$

so,

$$25, 9, 4, 1, 0, 121,$$

which sum to 160. Therefore,

$$s^2 = \frac{160}{6 - 1} = 32, \quad s = \sqrt{32} \approx 5.66.$$

- (c) State the units of the mean, IQR , variance, and standard deviation.

The mean, median, quartiles, IQR , range, and standard deviation are measured in dollars per hour. The variance is measured in dollars per hour squared.

- (d) Calculate the lower and upper $1.5IQR$ fences. Which observation, if any, is flagged? What does the flag establish, and what does it not establish?

The fences are

$$Q_1 - 1.5IQR = 6 - 4.5 = 1.5$$

and

$$Q_3 + 1.5IQR = 9 + 4.5 = 13.5.$$

The wage 20 lies above the upper fence and is flagged. The rule establishes its position relative to the quartiles and IQR under this convention. It does not establish that 20 is erroneous, impossible, generated by a different process, or suitable for deletion.

- (e) Replace 20 by 50. Calculate the new mean, median, and IQR ; determine whether the sample standard deviation increases or decreases; and use the result to explain resistance.

After replacing 20 by 50, the sum becomes 84 and the new mean is 14. The middle values and half-sample medians do not change, so $M = 7.5$, $Q_1 = 6$, $Q_3 = 9$, and $IQR = 3$. The new squared deviations from 14 sum to 1570, giving

$$s_{\text{new}}^2 = \frac{1570}{5} = 314, \quad s_{\text{new}} = \sqrt{314} \approx 17.72.$$

The median and IQR are unchanged, while the mean and standard deviation increase sharply. This illustrates resistance, but it does not mean the resistant summaries preserve the entire distribution.

- (f) Name two features of the original six observations that would be lost if someone reported only the mean and standard deviation.

Reporting only the mean and standard deviation loses, among other things, the ordered values, median and quartiles, the gap between 9 and 20, and the fact that one observation creates a long right tail. Any two such features answer the final part.

4. (From Practice 2, Question 5)

Two clinics record the following patient waiting times, in minutes:

A: 4, 5, 6, 7, 8, 9, 18

B: 4, 5, 6, 7, 8, 9, 48

- (a) Calculate the mean, median, Q_1 , Q_3 , and IQR for each clinic using the median-of-halves convention.

For Clinic A, the sum is 57 and

$$\bar{x}_A = \frac{57}{7} \approx 8.14.$$

For Clinic B, the sum is 87 and

$$\bar{x}_B = \frac{87}{7} \approx 12.43.$$

- (b) Without calculating both standard deviations exactly, determine which clinic has the larger standard deviation and justify your answer.

Both distributions have median 7. Excluding the median, each lower half is 4, 5, 6, so $Q_1 = 5$, and each upper half has median 9, so $Q_3 = 9$. Thus both have $IQR = 4$.

Clinic B has the larger standard deviation. Its largest value is 30 minutes farther to the right, which raises its mean and produces much larger squared deviations. An exact verification gives

$$s_A^2 = \frac{916}{42} \approx 21.81, \quad s_A \approx 4.67,$$

and

$$s_B^2 = \frac{10456}{42} \approx 248.95, \quad s_B \approx 15.78.$$

- (c) Apply the $1.5IQR$ rule to each distribution. Explain why receiving the same flag does not make the two largest waiting times equally unusual in magnitude.

For both clinics, the fences are

$$5 - 1.5(4) = -1$$

and

$$9 + 1.5(4) = 15.$$

Both 18 and 48 exceed 15 and are flagged. The flag is binary, but distance beyond the fence is not: 18 is 3 minutes above it, whereas 48 is 33 minutes above it. A graph of the original values therefore conveys information that a flag alone loses.

- (d) Write two sentences comparing the centers, spreads, and tails. Do not propose a cause for the difference.

Both clinics have median 7 minutes and $IQR = 4$ minutes, but Clinic B has a larger mean and standard deviation because its longest wait is much greater. Both maxima receive an upper-fence flag, while the original values and a graph reveal that Clinic B has the more extreme right tail.

Related Extra Practice

1. Observational studies suggest that countries with more generous unemployment benefits have higher rates of unemployment. If countries were to make unemployment benefits less generous, would this lead to lower unemployment rates? Discuss using relevant jargon.

We can't say for certain. The information given tells us that there is a positive correlation between unemployment benefits and rates of unemployment. However, this does not imply that changing unemployment benefits to be less generous will cause unemployment rates to decrease. There could be omitted/lurking variables that explain both unemployment benefits and unemployment rates at the national level. For example, a country whose labor force tends to work in highly volatile industries might both have high unemployment rates and pass high unemployment benefit laws.

2. Is the phone area code of a cell-phone owner a quantitative or qualitative variable? Explain. (The area code is a three digit number that is used to direct phone calls to a particular area of the U.S.)

The phone area code of a cell phone owner is a qualitative variable. Although the phone area code takes numerical values, they are essentially ID numbers. There is no meaningful way of arguing that one area code is larger than another, i.e., one cannot meaningfully order them.

3. Suppose a survey collects data on people's birth year, social security number and marital status. Which of these three variables are quantitative, and which are categorical? Explain.

Birth year is quantitative; social security number and marital status are categorical. Birth years can be meaningfully compared using operations of two real numbers ($<$, $>$, etc.) while values of the other two variables cannot.

4. You download a data file about student test scores, in which each row is a student's standardized test score for a particular subject (e.g., math, science, reading, history) in a particular school year. Students take more than one subject and we see their scores from more than one school year. What is the observational unit of the data file?

The unit is student-subject-year. We need all three of these elements to make a row unique.

5. Name two graphs that can be used to display the distribution of a categorical variable.

To display the distribution of a categorical variable, we could use a bar chart or a pie chart.

6. Construct a data set for $n = 6$ individuals of a quantitative variable such that the mean is 10, the median is 4, the first quartile is 2 and the third quartile is 7.

Let $x_1, x_2, x_3, x_4, x_5, x_6$ be the data points arranged in an ascending order. Then the first quartile is x_2 implying $x_2 = 2$, third quartile is x_5 implying $x_5 = 7$, median 4 implies $(x_3 + x_4)/2 = 4$, and mean 10 implies $(x_1 + x_2 + x_3 + x_4 + x_5 + x_6)/6 = 10$.

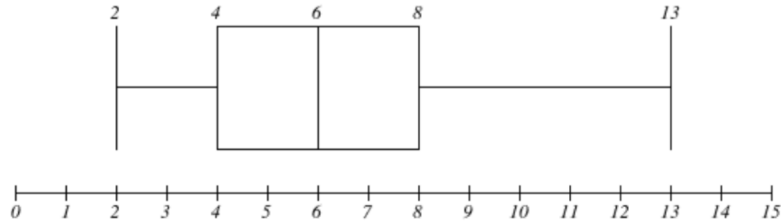
A solution to all these restrictions is

$$x_1 = 1, x_2 = 2, x_3 = 3, x_4 = 5, x_5 = 7, x_6 = 42.$$

Note that there can be many solutions.

7. Suppose a quantitative variable takes on the 6 values $\{1, 3, 3, 5, 7, 8\}$. Draw a boxplot of this distribution with numbers on the y-axis.
8. A quantitative variable x is observed for $n = 6$ individuals and takes on the values 2, 4, 5, 7, 8, 13.
 - Sketch the boxplot of this distribution.

We find the median of a data set with an even number of observations by taking the average of the middle two numbers in a sorted array. This yields a median of $0.5 * (5 + 7) = 6$. We find Q_1 by taking the median of the $n/2$ smallest data points, which gives us $Q_1 = 4$. Similarly, we find Q_3 by taking the median of the $n/2$ largest data points, which gives us $Q_3 = 8$. Therefore, our box plot takes the following shape:



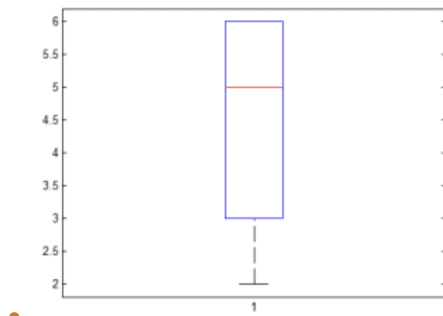
- Is 13 an outlier according to the $1.5IQR$ rule? Explain.
Data points that are more than 1.5 times the interquartile range [$IQR = 8 - 4 = 4$] larger than Q3 are considered outliers. This means points that are larger than $8 + 1.5 * 4 = 14$ are considered outliers in our case. Since 13 is smaller than 14 we conclude that 13 is not an outlier.
- Find another distribution of $n = 6$ values with the same median and interquartile range as x , but with mean equal to 20.
We can start with the same distribution and just adapt the largest value. This will neither affect the median nor the interquartile range. To find an outlier that pulls our mean up to 20 we solve the equation $(x + 2 + 4 + 5 + 7 + 8)/6 = 20$. This yields $x = 94$. Therefore one possible solution to the question is the distribution $[2, 4, 5, 7, 8, 94]$.

9. The distribution of the variable x for $n = 50$ individuals is as follows

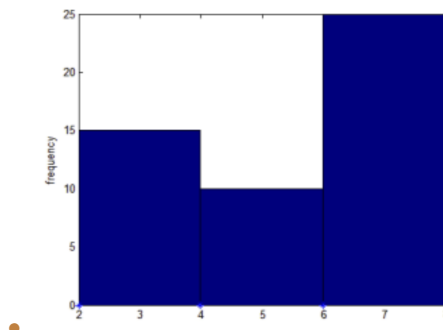
x	6	3	4	2
frequency	25	13	10	2

that is, $x = 6$ for 25 individuals, $x = 3$ for 13 individuals, etc.

- (a) Sketch a boxplot of this distribution, with numbers on the y-axis.



- (b) Sketch a histogram of this distribution, with the first bin starting at 2, and bin size equal to 2. Indicate “frequencies” on the y-axis.



- (c) Consider the variable $y = -2x + 7$, that is for each individual i with x -value equal to x_i , $y_i = -2x_i + 7$. What fraction of the y values is larger than zero? Explain.

Fraction of y values that is larger than 0 is the same as fraction of x values that is smaller than 3.5 and hence the answer is 0.3

10. A quantitative variable measured for $n = 40$ individuals takes on the values -2 for 10 individuals, the value 4 for 29 individuals, and the value 12 for the remaining individual.

- Compute the mean and standard deviation of this distribution.

$$\begin{aligned}\bar{x} &\equiv \frac{1}{40} \sum_{i=1}^{40} x_i = \frac{1}{40} [-2 \times 10 + 4 \times 29 + 12 \times (40 - 10 - 29)] \\ &= 2.7\end{aligned}$$

$$\begin{aligned}s_x &\equiv \sqrt{\frac{1}{39} \sum_{i=1}^{40} (x_i - \bar{x})^2} \\ &= \sqrt{\frac{1}{39} [10(-2 - 2.7)^2 + 29(4 - 2.7)^2 + 12(12 - 2.7)^2]} \\ &\approx 3.02\end{aligned}$$

- Is the value 12 an outlier of this distribution according the rule-of-thumb for outliers? Explain.

$$\begin{aligned}\text{Median} &= 4 \\ Q1 &= \frac{(-2 + 4)}{2} = 1 \\ Q3 &= 4 \\ IQR &= 3 \\ 12 - 4 &= 8 > 1.5 \times 3 = 4.5\end{aligned}$$

So yes, the value 12 is an outlier in this distribution because it is more than 1.5 IQR above Q3.

11. The members of a basketball team are attempting 20 free throws each. Let x_i denote the number of successful attempts for player i . Suppose the mean of x , $\bar{x}$, equals 12.6 and the sample standard deviation, s_x , equals 4.2. Let y_i be the number of failed attempts of player i .

- What are the mean and standard deviation of y_i ?

Since each player attempts 20 free throws,

$$y_i = 20 - x_i \tag{1}$$

Using what we know about means and standard deviations,

$$\begin{aligned}\bar{y} &= 20 - \bar{x} \\ &= 7.4\end{aligned} \tag{2}$$

and

$$\begin{aligned}s_y^2 &= s^2(20 - x) \\ &= (-1)^2 s_x^2 \\ &= s_x^2\end{aligned} \tag{3}$$

hence

$$\begin{aligned}s_y &= s_x \\ &= 4.2\end{aligned}\tag{4}$$

We could also write out the variance formula explicitly:

$$\begin{aligned}s_y^2 &= \frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2 \\ &= \frac{1}{n-1} \sum_{i=1}^n (20 - x_i - (20 - \bar{x}))^2 \\ &= \frac{1}{n-1} \sum_{i=1}^n (\bar{x} - x_i)^2 \\ &= s_x^2\end{aligned}\tag{5}$$

- How would a scatter plot of the variables x and y look like? Be as specific as you can.

Each point would lie on the line $y = 20 - x$.

12. Suppose a data set records the income of individuals. There are $n_1 = 153$ records on women, and the average income of women is \$36,000. There are also $n_2 = 210$ records on men, and the men's average income is \$38,000. What is the average income of all of the $n_1 + n_2$ individuals?

Since the average income of the $n_1 = 153$ women is \$36,000, the sum of the incomes of these 153 women is $153 \times \$36,000 = \$5,508,000$. Similarly, the sum of the incomes of the $n_2 = 210$ men is \$7,980,000. Therefore the average income of all the $n_1 + n_2 = 363$ individuals is:

$$\frac{5508000 + 7980000}{363} = \$37,157.02$$